

Nirajan Paudel

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EDUCATION

University of Colorado Boulder Master of Science in Computer Science	Boulder, CO May 2027
Institute of Engineering, Tribhuwan University Bachelor of Engineering in Electronics, Communication & Information Engineering	Pokhara, Nepal 2020–2024

WORK EXPERIENCE

Graduate Student Researcher <i>LECS lab, University of Colorado Boulder</i>	Oct 2025 – Present
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- Investigated speculative decoding across 11 low-resource languages and 2 model families (Qwen 3.5, Llama 3.2), showing draft/verifier divergence and acceptance rates correlate with language resourcedness ([paper under review](#), EMNLP).
- Designed task-specific and general-domain distillation pipelines to reduce draft/verifier divergence, uncovering poor cross-task generalization for low-resource languages.
- Proposed n-gram draft models as a low-cost alternative to neural drafters, achieving superior speed-up factors across all languages.

Machine Learning Intern <i>PalmMind Technology</i>	March 2025 – May 2025
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- Architected production agentic RAG chatbots for EV dealers and insurance clients using OpenAI GPT-4, migrating from LangChain to LangGraph to enable advanced tool calling and parallel processing with orchestrator-worker patterns.
- Implemented LLM routing strategy (GPT-4 for complex queries, GPT-3.5-turbo for classification), reducing API costs while maintaining quality through prompt engineering and Redis caching for 500+ customer inquiries.
- Optimized retrieval pipeline integrating web scraping, Tesseract OCR, and context management, improving accuracy for insurance policy and vehicle specification queries.

Teaching Assistant <i>Pashchimanchal Campus, Tribhuwan University</i>	Sep 2024 – Feb 2025
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- Instructed 90+ students in C programming and Information Systems, designing lab exercises covering cloud computing, distributed systems, neural networks, and data mining.
- Mentored 15+ semester projects, achieving 85% student proficiency in algorithm design and implementation.

PROJECTS

Tensor AI - Context-Aware Academic Assistant  <i>Personal Project</i>	Nov 2025 – Present
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- Architected production-grade RAG pipeline using FastAPI and Pinecone vector database, processing 10+ engineering programs with multimodal ingestion through LlamaParse for legacy PDFs containing technical diagrams and tables.
- Implemented agentic query routing with intent classification and cross-encoder reranking (ms-marco-MiniLM-L-6-v2), achieving 95% retrieval relevance through dynamic top_k scaling (3-50 chunks) based on query complexity.
- Optimized system performance with 7-layer Redis caching strategy for embeddings (7-day TTL) and LLM classifications (24-hour TTL), reducing API costs by 70% and achieving sub-200ms initial response time using Server-Sent Events streaming.
- Built React frontend with real-time SSE streaming and automated semester detection based on Nepali calendar (B.S.) for personalized, context-aware academic support without manual user configuration.

Serverless vs Kubernetes NLP Inference Benchmark  <i>Course Project, University of Colorado Boulder</i>	Oct 2025 – Dec 2025
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- Architected comparative cloud infrastructure deploying DistilBERT sentiment analysis on AWS Lambda (serverless) and EKS (Kubernetes), evaluating latency, cost, and scalability under variable load conditions with automated Locust testing.
- Engineered full-stack deployment pipeline using Terraform IaC for VPC, EKS cluster, Lambda functions, and API Gateway, with GitHub Actions CI/CD pushing containerized services to ECR and orchestrating multi-environment deployments.
- Built Streamlit dashboard with FastAPI backend for real-time parallel inference comparison, implementing exponential backoff to handle Lambda cold starts (60s model loading) and integrating LLM-powered performance analysis for SRE insights.
- Optimized container orchestration with Kubernetes HPA (Horizontal Pod Autoscaler) and LoadBalancer services, managing resource constraints (6Gi RAM limits) while maintaining service reliability across 2-replica deployments.

TECHNICAL SKILLS

Languages	Python, C/C++, Java, SQL, Bash
ML/AI	PyTorch, TensorFlow, LangChain, LangGraph, HuggingFace, OpenAI/Anthropic APIs, RAG, Transformers
Cloud & DevOps	AWS (Lambda, EKS, EC2, ECR), GCP (GKE, GCS), Kubernetes, Docker, Terraform, GitHub Actions
Tools & Databases	FastAPI, Flask, Redis, Pinecone, PostgreSQL, Tree-sitter, Git, Pandas, NumPy